OPERATING SYSTEMS – ASSIGNMENT 1

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GitHub Repository:

TASK 2 – Judge the Type of Operating System for Each Environment (3 Marks)

Compare mainframe, desktop, multiprocessor, distributed, clustered, real-time (hard and soft) and handheld systems.

Discuss which type best fits monitors, tablets, workstations, database server, imaging server and partner link; symmetric versus asymmetric multiprocessing for the imaging server; and a clustered system with fail-over for the database server.

Alternative Concept and Complexity Analysis: If the database server used one large multiprocessor machine instead of a cluster of smaller machines with fail-over, evaluate availability, cost, throughput growth with processors and memory available.

Evidence Expected: A justified choice of OS type for each environment, with rejected alternatives and reasons.

ANSWER

2.1 Selection by environment

Hospital component	Recommended OS type	Rejected alternatives and justification
Bedside monitors	Embedded hard real-time OS (for example, QNX or a suitably configured RTOS)	Desktop/time-sharing OSs cannot by themselves guarantee a 10 ms deadline. A mainframe is unnecessary and network dependence is unsafe.
Nurses' tablets	Handheld/mobile OS (for example, Android Enterprise or managed iPadOS)	Desktop OSs consume more power and are less convenient for touch-first mobile use.
Doctors' workstations	Desktop OS (managed Windows or Linux)	Handheld OSs may not support specialist workstation applications and large displays as effectively.
Patient-record database	Clustered server OS with fail-over and replicated storage	One standalone server is a single point of failure. A mainframe may offer consolidation but does not remove all single-machine failure risks.
MRI imaging server	Multiprocessor server OS, preferably Linux; symmetric multiprocessing (SMP)	A single-processor system limits throughput. Asymmetric multiprocessing can be useful for fixed specialised roles but is less flexible for general parallel image workloads.
Partner-hospital record sharing	Distributed system using secure network services and interoperable APIs	A single central machine creates distance, availability and ownership bottlenecks. A simple cluster inside one site does not alone solve cross-site distribution.

2.2 Mainframe, multiprocessor, distributed, clustered and real-time concepts

- Mainframe systems are designed for high-volume, centralised transaction processing and many concurrent users. Their reliability and throughput can support large enterprise workloads, but a mainframe is not required for every endpoint.
- Desktop systems prioritise interactive use, broad peripheral support and application compatibility.
- Multiprocessor systems contain multiple CPUs/cores sharing memory and I/O resources. They improve parallel workload throughput, subject to memory bandwidth and synchronisation limits.

- Distributed systems consist of networked computers that coordinate and share services. They are suitable for three hospitals and cloud records, but network partitions and latency must be handled.
- Clustered systems are multiple computers configured to provide a service together. In an active-passive database cluster, a standby node can take over when the active node fails. Shared or replicated storage and quorum/fencing are needed to avoid split-brain.
- Hard real-time systems must meet deadlines; missing a deadline is considered system failure. Soft real-time systems prefer deadlines but occasional misses degrade quality rather than cause immediate catastrophic failure. The monitor alarm is hard real-time; MRI display or image-processing completion is generally soft real-time.

2.3 SMP versus AMP for MRI

SMP gives each processor equal access to the operating system and shared memory. The scheduler can distribute MRI reconstruction threads across available cores, improving load balancing and flexibility. AMP assigns processors fixed or specialised roles, which can offer predictable execution for dedicated pipelines but is less flexible and can leave processors underused. For a general MRI reconstruction server, SMP is preferred because the workload contains parallel tasks and needs dynamic load balancing.

2.4 Alternative: one large multiprocessor database server instead of a cluster

A large multiprocessor server can deliver high throughput by running many database workers in parallel, but it remains a single point of failure: motherboard, power, kernel, memory-controller or maintenance failure can stop the whole database. A cluster of smaller machines with automatic fail-over can restore service within seconds, depending on detection, fencing, recovery and application retry time. A cluster costs more in networking, replication, licences and administration, but improves availability.

Amdahl's Law explains processor scaling: if fraction P of a workload is parallel and there are k processors, ideal speedup is $S(k) = 1 / ((1 - P) + P/k)$. Thus throughput rises with k but has diminishing returns as serial work, locking and I/O become bottlenecks. Memory capacity is approximately the sum of usable RAM across nodes only when the application can distribute its data; a single server's RAM is limited by its hardware. Cluster nodes do not automatically behave as one shared-memory machine. Space complexity of stored patient data is $O(D)$, where D is total record data; replication may require roughly rD storage for r copies.

QUESTION FROM THE PROVIDED ASSIGNMENT